

Photoresist Contrast-Curve Characterization (AZ1512)

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Objective

Measure the exposure–development response of AZ1512 positive photoresist and extract the two figures of merit that define a lithography process window — the resist contrast γ and the dose-to-clear E_0 — together with wafer-level coat uniformity and wafer-to-wafer dose variation.

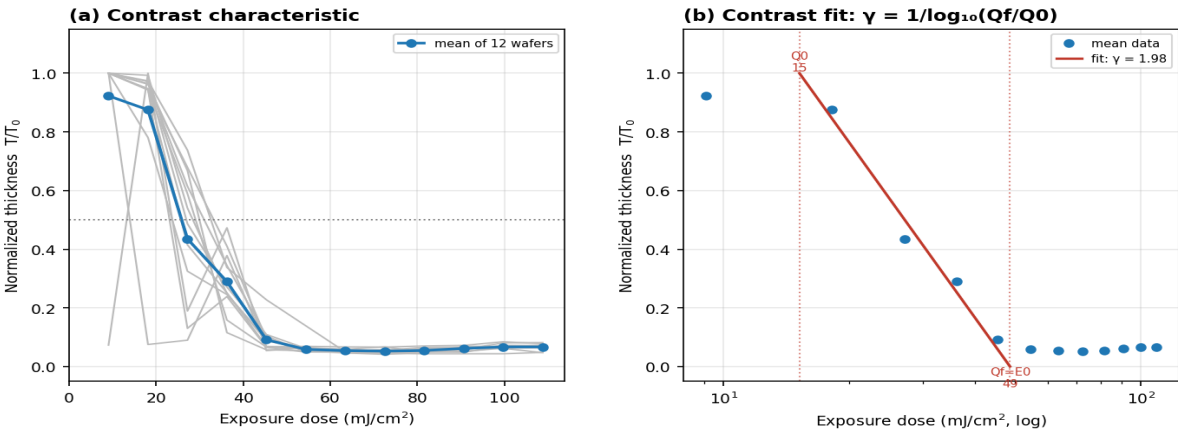
Method

AZ1512 was spin-coated on SiO_2/Si and soft-baked, then exposed on a contact aligner at $32 \text{ mW}/\text{cm}^2$ in 12 evenly spaced dose steps (longest = $3.4 \text{ s} = 108.8 \text{ mJ}/\text{cm}^2$), replicated across 12 wafers. After TMAH development, remaining resist thickness was measured by Filmetrics reflectometry (227 of 229 fits valid, median goodness-of-fit 0.99). A Python pipeline pulls the metrology export directly from Google Drive, rejects low-quality fits, normalizes each wafer to its own unexposed thickness T_0 , and fits normalized thickness against $\log_{10}(\text{dose})$ over the clearing transition, giving $\gamma = -\text{slope} = 1 / \log_{10}(Q_f/Q_0)$.

Results

Parameter	Value	Meaning
Contrast γ	1.98	slope of the clearing transition
Onset dose Q_0	$15.2 \text{ mJ}/\text{cm}^2$	extrapolated full-thickness intercept
Dose-to-clear E_0 (Q_f)	$48.6 \text{ mJ}/\text{cm}^2$	dose that fully clears the film
Wafer-to-wafer D_{50}	$26.8 \pm 5.1 \text{ mJ}/\text{cm}^2$	dose at 50% remaining (19% spread)
Coat uniformity T_0	$1.091 \pm 0.062 \mu\text{m}$	5.7% 1-sigma across 12 wafers
Exposure margin	$E_0 = 45\%$ of working dose	$\sim 2.2\times$ headroom at $108.8 \text{ mJ}/\text{cm}^2$

Photoresist Contrast Curve & Parameter Extraction (AZ1512)



(a) Normalized resist thickness vs. exposure dose for all 12 wafers (mean in blue). (b) Semilog fit of the clearing transition yielding γ , with onset Q_0 and dose-to-clear Q_f marked.

Discussion

The measured response is textbook positive-resist behavior: full film retained at low dose, a steep clearing transition, and a residual floor once cleared. A contrast of $\gamma \approx 2$ is consistent with broadband contact-aligner exposure of AZ1512. With E_0 at $\sim 45\%$ of the production dose, the process holds a comfortable $\sim 2.2\times$ exposure margin, so the working setpoint is in no danger of under-clearing. The 5.7% coat and 19% dose-to-clear spreads quantify the real process variation and frame the statistical-process-control expectations for critical-dimension control in high-volume lithography. The full analysis is reproducible from the committed data and script.